Gaurav Niranjan

AI Engineer · Tübingen, Germany · open to relocation anywhere in Germany
Post-study job-seeker permit · no sponsorship required
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SELECTED PROJECTS

Income Tax Agent: RAG assistant for India's Income-tax Act, 2025 (live demo) *2026*

- Built a **LangGraph** pipeline that answers questions about a 674-page legal act **only from retrieved text**, with section citations. Stack: Gemini API, Chroma, local BGE embeddings, BM25, Streamlit app.
- Found a root cause of wrong answers: amendment footnotes quote **repealed law**. Wrote a PDF cleaner that strips them and their in-text markers across 553 printouts, tested on real page snippets.
- Built an **eval harness** with versioned run configs and used it for every design choice: dense vs. BM25 vs. hybrid (RRF), section-title chunk prefixes, neighbour chunks and **LLM query expansion**. Right section at rank 1 rose from **13/22 to 18/22**, MRR from **0.69 to 0.88** (**0.88** on a held-out set). LLM judge: 17/18 answers faithful to the sources.
- Test suite with **pytest** and **GitHub Actions CI**, using a fake LLM and fake embeddings so it runs without API keys.

Biogeography-Based Feature Selection for Hyperspectral Images *Feb 2023 – Oct 2023*

- Externally funded by the **Indian Space Research Organisation** (Regional Academic Centre for Space). Built a band-selection pipeline with Biogeography-Based Optimization to pick the most informative spectral bands of hyperspectral images.
- Reached **96.68% accuracy** (κ 0.95, avg. F1 0.94) on Indian Pines, beating Genetic Algorithm (93.07%) and Diff. Evolution (95.02%). Published at *CVIP 2023*.
- Extended with CNNs in a **Springer book chapter** (2025): **86%** fewer 3D-CNN parameters and 69% faster training.

MASTER'S THESIS

Text and Image Contributions to Gender Bias in VLMs *Jan 2026 – Jul 2026*

- Benchmarked **10 open VLMs** (Qwen, Gemma, LLaVA) on gender-bias benchmarks with **interpretability methods**, using **vLLM** for fast forward passes on a Slurm GPU cluster.
- Built a diffusion (FLUX.2) pipeline that swaps image backgrounds while keeping the subject fixed, to isolate the background's causal effect. Designed a patch-ablation method with PyTorch hooks to find where bias starts in an image, including Qwen's DeepStack architecture.
- Logistic regression: question wording shifts the odds of a male answer **2 to 4×**; background edits had a small opposite effect.

WORK EXPERIENCE

Student Research Assistant — Tübingen AI Center, University of Tübingen *Oct 2025 – Dec 2025*

- Computer Vision and Multimodal Learning Group. Built an ingestion pipeline that turned two YouTube video-caption datasets (**600 GB**) into sharded WebDataset format, aligning frames with annotations. Published both on the Hugging Face Hub.

Student Research Assistant — Chair of Marketing, University of Tübingen *Oct 2024 – Sep 2025*

- Fine-tuned diffusion models with **DreamBooth and LoRA** to generate apparel images with target attributes; studied the structure and interpretability of generative-model weight space for fashion-trend analysis.

Research Intern — National Institute of Technology Warangal, India (remote) *May 2022 – Jan 2023*

- Implemented the first prototype of Deep MMMF, a deep extension of Maximum Margin Matrix Factorization. Final model: up to **11% lower error** (NMAE) than the best of 13 baselines (Friedman/Nemenyi tests). Published in *Scientific Reports*.

EDUCATION

M.Sc. Machine Learning — University of Tübingen · final grade **1.8** (1.0 = best) *Oct 2023 – Aug 2026*

Coursework: Deep Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Recurrent & Generative Neural Networks, Video Analytics, Statistical & Probabilistic ML.

B.Tech. Computer Science — LNM Institute of Information Technology, Jaipur · CGPA **9.47 / 10** *Aug 2019 – Jun 2023*

PUBLICATIONS

- S. Kumar, V. R. Kagita, V. Kumar, **G. Niranjan**. Deep Maximum Margin Matrix Factorization. *Scientific Reports* 16, 14518 (2026).
- A. Datta, **G. Niranjan**, S. Ghosh, A. Ghosh. Optimizing Hyperspectral Image Classification: Biogeography-Based Band Selection and CNN Analysis. *Intelligent Systems Reference Library* 268, Springer, 2025.
- A. Datta, **G. Niranjan**. Biogeography Based Band Selection for Hyperspectral Image Classification. *CVIP 2023*, Springer, 2024.

TECHNICAL SKILLS

LLMs & retrieval: Generative AI, Large Language Models (LLMs), NLP, LangGraph, LangChain, Gemini API, OpenAI API, vLLM, RAG, Chroma, Pinecone, semantic search, hybrid retrieval, embeddings, prompt design, evals, LLM-as-judge
ML: Computer Vision, Vision-Language Models (VLMs), CNNs, PyTorch, Hugging Face Transformers, scikit-learn, LoRA / DreamBooth fine-tuning, interpretability (hooks, ablation), NumPy, Pandas
Engineering: Python, SQL, C++, Java, Bash, pytest, GitHub Actions, Git, Docker, Linux, Slurm / HPC, AWS, Google Cloud, Streamlit
Languages: English (C1), German (B1, working toward B2), Hindi (native)